



Involvement of the SCO3366 efflux pump from *S. coelicolor* in rifampicin resistance and its regulation by a TetR regulator

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Abstract

Overexpression of efflux pumps represents a key mechanism of resistance in bacteria. Soil bacteria such as *Streptomyces* harbour a vast array of efflux genes that are transcriptionally silent under laboratory conditions. However, dissemination of many of these genes into clinical pathogens via horizontal gene transfer results in conferring resistance to multiple drugs. In this study, we have identified the role of a MFS transporter, SCO3366 from *Streptomyces coelicolor*, in governing multidrug resistance. Overexpression and knockout studies revealed that SCO3366 provides resistance to several structurally unrelated drugs including ciprofloxacin, chloramphenicol, rifampicin and EtBr, with rifampicin being the major substrate. Beyond multidrug resistance, SCO3366 was efficient in providing tolerance towards oxidative stress. A combinatorial mechanism of increased oxidative stress tolerance decreased intracellular drug levels and decreased permeability act synergistically to provide resistance towards rifampicin. Shedding light on the regulation of SCO3366, we find the pump to be directly regulated by the TetR regulator SCO3367 in a negative manner and the repression was found to be relieved in presence of different compounds recognized as substrates of SCO3366.

Key points

- First reported rifampicin efflux pump in *Streptomyces coelicolor*
- Resistance to rifampicin is the result of a synergistic action of increased efflux with increased oxidative stress tolerance and decreased permeability, which can potentially arise in clinically relevant bacteria
- SCO3366-SCO3367 to be a novel system that operates to protect the bacteria under varied environmental stress conditions

Keywords Antibiotic resistance · Rifampicin resistance · Oxidative stress · Solvent stress · Efflux pumps · MFS transporter · *Streptomyces coelicolor*

Introduction

Active efflux is one of the efficient ways by which bacteria gain resistance to multiple drugs, thus allowing them to survive under decreased intracellular drug levels (Li and Nikaido 2009; Webber and Piddock 2003). These efflux pumps can be specific for a single substrate or can expel multiple substrates, thus conferring multidrug resistance phenotype. Efflux pumps are classified as either primary transporters, which require energy from ATP hydrolysis to drive the efflux, or as secondary transporters, which utilize

a proton-motive force to efflux out the drugs. It is estimated that 5–10% of all bacterial genomes are composed of transport proteins, a major fraction of which encode the efflux pumps (Saier and Paulsen 2001). AcrAB in *E. coli*, MexAB-OprM in *Pseudomonas aeruginosa*, NorA in *Staphylococcus aureus* and PmrA in *Streptococcus pneumoniae* are few of the extensively studied efflux pumps that confer resistance to multiple drugs (Costa et al. 2019; Llanes et al. 2004; Nikaido and Zgurskaya 2001). Apart from providing resistance to different drugs, efflux pumps also operate to provide tolerance to oxidative stress, solvent stress, metal toxicity and virulence in different organisms (Piddock 2006; Piddock et al. 2002). For example, the TtgABC and TtgGHI efflux pumps in *Pseudomonas putida* DOT-T1E operates to provide tolerance to the solvent toluene (Fraud and Poole 2011; Rodríguez-Herva et al. 2007). P55 and CmeABC are important

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efflux pumps that provide tolerance to oxidative stress in *Mycobacterium tuberculosis* and *Campylobacter jejuni* respectively (Lin et al. 2002; Machado et al. 2017; Ramón-García et al. 2009). The MacAB efflux pump in *Salmonella typhimurium* has been found to be essential in virulence and oral infection in mice (Bogomolnaya et al. 2013). Interestingly, overexpression of efflux pumps combines synergistically with other cellular factors, thus leading to increased drug resistance. For example, overexpression of the efflux pump MexXY induces tolerance to oxidative stress, thus leading to enhancement of aminoglycoside resistance in *Pseudomonas aeruginosa* (Fraud and Poole 2011). Efflux pump overexpression also synergizes with outer membrane barrier in different Gram-negative bacteria, thus leading to elevated resistance to different drugs (Zgurskaya et al. 2018).

Streptomyces are Gram-positive bacteria that produce 60% of the world's total antibiotics. In order to gain resistance to the endogenous antibiotics, a self-resistance mechanism has been adopted by these organisms, which is predominantly efflux mediated (Mak et al. 2014; Wang et al. 2017). Apart from the genes required for self-resistance, *Streptomyces* also carry efflux genes gained from other actinobacteria through horizontal gene transfer (Egan et al. 2001). Thus, many *Streptomyces* also harbour resistance genes for antibiotics which are not endogenously produced by them (D'Costa et al. 2006). For example, vancomycin resistance gene has been found to be present in *Streptomyces coelicolor*, a non-producer of the antibiotic (Hong et al. 2004). In certain cases, many of these resistance genes are found to be disseminated to different pathogenic bacteria through several horizontal gene transfer routes, thus conferring resistance to multiple drugs (Jiang et al. 2017).

About 10% of the *Streptomyces* genome is devoted to transporters. Some of them are importers such as the RibXY in *Streptomyces davaonensis* and Ceb/MsiK in *Streptomyces scabies*, which are essential for uptake of riboflavin and cello-oligosaccharide respectively (Jourdan et al. 2016; Schneider and Mack 2021), whereas others are exporters, that are involved in export of antibiotics, metals, toxic metabolites, etc. The latter are efflux pumps that constitute about 15% of the total transporter pool (Zhou et al. 2016). Some of the widely studied efflux pumps in *Streptomyces* include the ABC transporters DrrAB in *Streptomyces peucetius* (Li et al. 2014) and OtrC in *Streptomyces rimosus*, which provide resistance to other exogenous drugs, besides providing resistance to the endogenous antibiotics daunorubicin and oxytetracycline respectively (Yu et al. 2012). Earlier studies have recognized the role of two efflux pumps CmlR1 and CmlR2 in *Streptomyces coelicolor* that provide resistance to the exogenous antibiotic chloramphenicol, with no role in endogenous antibiotic resistance (Vecchione et al. 2009). In a recent study, we have characterized an efflux pump SCO4121, belonging to Major Facilitator Superfamily

(MFS) of transporter proteins, from *Streptomyces coelicolor* to be involved in resistance to multiple exogenous drugs and the oxidant HOCl (Nag and Mehra 2021).

One of the effective mechanisms to understand the transcriptional activity of an efflux pump is via subjection to different stress (Li et al. 2018). In a previous transcriptomic study in *Streptomyces coelicolor*, we have identified a large number of efflux pumps to be involved in increased resistance towards the fluoroquinolone, ciprofloxacin (Patkari et al. 2018; Patkari and Mehra 2013). From this pool of transporters, we identified a widely conserved, MFS pump, SCO3366, which was upregulated in response to ciprofloxacin in both nutrient-rich and nutrient-deficient conditions. Besides, upregulation of *sco3366* has also been reported in response to acidic pH conditions (Kim et al. 2008). This information thus suggested SCO3366 to be transcriptionally active under differential stress environments. In this work, through overexpression and knockout studies, we delineate the role of SCO3366 in conferring multidrug resistance and tolerance to oxidative stress. Furthermore, through promoter activity assays, we identified a TetR regulator SCO3367 to be involved in negative regulation of SCO3366 by sensing a broad spectrum of inducers that possibly interacts with the repressor molecule, thus affecting the DNA binding activity of the repressor and allowing efficient transcription of *sco3366*.

Materials and methods

Bacterial growth procedures

Streptomyces coelicolor (WT86 and recombinants) spores ($\sim 10^8$ spores) were germinated in 2XYT media (2% yeast extract, 1% NaCl, 1% tryptone). The germinated mycelia was further transferred to R5 media (1% w/v of $MgCl_2$, 10% sucrose, 1% glucose, 0.003% K_2SO_4 , 0.002% CAS amino acids, 0.5% yeast extract, 0.5% TES buffer). Samples were taken at different intervals, homogenized by brief sonication and O.D monitored by measuring absorbance at 450 nm (Bradshaw et al. 2013). Cells corresponding to 1 O.D were used for all molecular and biochemical assays. For generation of knockout strains, 39 °C was the chosen temperature. *E. coli* Dh5 α cells were used as the cloning host and grown in Luria Bertaini (LB) broth at 37 °C for all growth conditions. *Mycobacterium smegmatis* MC²155 cells were grown in LB media supplemented with 0.05% of Tween 80. Mannitol Soya Agar Plates (mannitol-2%, soya-2% and agar-2%) were used for generation of spores and for carrying out the phenotypic assays. Apramycin at a concentration of 50 μ g/ml was used for selection of *E. coli* recombinants,

whereas 25 µg/ml was the concentration for selection of *S. coelicolor* recombinants.

Generation of *sco3366* overexpression, knockout and complementation strains

For overexpression of *sco3366* into *S. coelicolor* cells, the gene was amplified using specific primers mentioned in Table S1. The amplified PCR product was further digested with the enzymes *Bam*HI and *Hind*III and inserted into the multi-copy plasmid pIJ86 digested with the same enzymes, to form the directional clone O3366. For yielding the *sco3366* deleted strain Δ3366, the left (LFR) and right flanking (RFR) regions corresponding to respective gene fragments were amplified by PCR from the genomic DNA of WT *S. coelicolor* with primers listed in Table S1. The LFR fragment was digested with the enzymes *Eco*RI and *Bam*HI and RFR fragment was digested with the enzymes *Xba*I and *Hind*III. These digested LFR and RFR fragments were cloned into the deletion plasmid pKC1139. The constructs were transformed into the conjugative *E. coli* strain ET12567 (pUZ8002). These constructs were plated onto MSA agar plates consisting of 10 mM MgCl₂. After incubation at 30 °C for 20 h, these plates were flooded with apramycin (1mg/ml) and nalidixic acid (500 µg/ml). The plates were further incubated at 30°C for 28 h till pin-sized colonies were obtained, further which the temperature was shifted to 39°C. pKC1139 is an *E. coli-Streptomyces* shuttle plasmid having a temperature-sensitive origin of replication, and therefore unable to grow beyond 34°C. Thus, the colonies obtained at 39°C will be those formed by single crossover events. For obtaining double crossover mutants, the cells were allowed to undergo three rounds of sporulation in the absence of any antibiotic. Furthermore, to yield the complementation strain cΔ3366, *sco3366* cloned into the pIJ86 plasmid was transformed into the Δ3366 strain. O3366, Δ3366 and cΔ3366 cells were further confirmed by plating onto R5 plates with 25 µg/ml of apramycin and also through PCR amplification and sequencing.

Preparation of *sco3366* promoter constructs

To prepare the *sco3366* promoter construct, *sco3366* promoter corresponding to 155bp located upstream to the translational start site was amplified from *S. coelicolor* genomic DNA, digested with enzymes *Eco*RI and *Bam*HI and placed upstream of pFpV27 plasmid to form the construct P₃₃₆₆::*gfp*. Separately, *sco3367*, amplified from *S. coelicolor*, was cloned into the integrative plasmid pST-ki under the restriction sites *Eco*RI and *Hind*III, and transformed into *Mycobacterium smegmatis* via electroporation allowing its constitutive expression and yielding the strain MS3367. P₃₃₆₆::*gfp* construct was transformed

into previously prepared competent MS3367 cells via electroporation to yield the strain MS3367::P₃₃₆₆·P₃₃₆₆::*gfp* was transformed into WT *M. smegmatis* cells to form the construct P₃₃₆₆, cells which were devoid of SCO3367.

Gene expression studies

Bacterial cells (both *S. coelicolor* and *M. smegmatis*) were grown till OD reached ~1. Cells were then treated with 12% of 5% phenol-ethanol mixture, centrifuged at 8000 rpm for 10 min and stored at –80°C. Total RNA from these cells was extracted using Trizol method as per defined protocols (Maarsingh et al. 2019; Van Dessel et al. 2004). DNA from the extracted RNA was removed using 50 U of DNase (Invitrogen). cDNA synthesis was carried out using 4 µg of template RNA using Revert Aid H Minus Reverse Transcriptase (Thermo). Reaction mixture for cDNA synthesis included 4 µg of RNA, 1X Reaction buffer, 0.2 µg random primers, 0.2 mM dNTPs and 1 U RT enzyme in a total reaction volume of 20 µl. cDNA synthesis was carried out by using the following reaction steps: 25°C for 10 min, 50°C for 1 h and deactivation at 72°C for 10 min. cDNA was further diluted to a final concentration of 100 ng/µl after synthesis. Expression studies were carried out using 100 ng of cDNA template and 0.5 µM of specific primers (listed in Table S2). The reaction was carried out using iTaq Mix (Biorad) in Biorad CFX96 real-time PCR. Data analysis was performed using 2^{–ΔΔCt} method. Normalization was carried out using the house-keeping gene and further by the untreated sample. 23S rRNA was used as the house-keeping gene for *S. coelicolor*, whereas 16S rRNA was used as the house-keeping gene for *M. smegmatis* cells for all expression studies. The final data was plotted as per defined protocols (Schmittgen and Livak 2008).

Antimicrobial susceptibility assay

Disc diffusion assay

For measuring drug sensitivity via disc diffusion technique, a modified protocol was adopted as previously discussed (Keenan et al. 2019). 10⁸ spores of the specific *S. coelicolor* strains (WT86 and recombinants) were streaked onto R5 plates with 25 µg/ml apramycin. These plates were further incubated for 72 h at 30°C. Zone of inhibition around the disc was measured (in millimetres) after the incubation period. Higher the zone, greater was the susceptibility against the particular drug. Susceptibility of all *S. coelicolor* strains (listed in Table S3) was tested using this assay. Different drugs used for the assay include chloramphenicol, ciprofloxacin, norfloxacin, ofloxacin, streptomycin, EtBr, ampicillin, apramycin, erythromycin, kanamycin and CCCP. All drugs were purchased from Himedia except apramycin and ciprofloxacin, which were purchased from Sigma-Aldrich.

ZnSO₄, FeSO₄, CoSO₄ and CuSO₄ were the redox salts towards which O3366 and Δ3366 cells were screened. All metal salts were purchased from Himedia.

Agar dilution method

R5 agar plates were prepared with desired concentrations of the antibiotics to be tested. 10⁸ spores of *S. coelicolor* were streaked onto these plates. These plates were further incubated at 30 °C for 72 h. Growth was checked after the incubation period. The antibiotic concentration at which growth inhibition was observed was considered to be the MIC of the particular drug. Different drugs used for the assay include chloramphenicol, ciprofloxacin, norfloxacin, ofloxacin, streptomycin, EtBr, ampicillin, apramycin, erythromycin, kanamycin and CCCP. All drugs were purchased from Himedia except apramycin and ciprofloxacin, which were purchased from Sigma-Aldrich.

Spot dilution assay

S. coelicolor cells WT86 and recombinants were grown till 1 O.D. 10⁸ cells were taken into 24-well plates and 2 ml of R5 media was added. The cells were further treated with desired conditions and incubated for 72 h. Cells were further diluted 100 times and 10 µl of the culture were spotted on R5 plates. Growth was monitored after 72 h of incubation at 30°C.

Membrane permeability assay

Membrane permeability was measured using the fluorescent dye, NPN (1-Phenyl-naphthalmine) (Helander and Mattila-Sandholm 2000). NPN is a hydrophobic dye that is excluded from the intact membrane. However, upon permeabilization, NPN diffuses into the cells, and binds to the hydrophobic membrane, thus leading to increased fluorescence. *S. coelicolor* cells (WT86 and recombinants) were grown till 1 O.D at 30°C. The cells were harvested at 12,000 rpm for 10 min at 4°C. The cells were then resuspended in 5 mM 2-[4-(2-hydroxyethyl) piperazin-1-yl] ethanesulfonic acid (HEPES) buffer. One hundred microlitres of the sample along with 100 µl of 50 µM NPN prepared in HEPES buffer was taken in 96-well plate and fluorescence was measured immediately at 355 nm excitation and 402 nm emission wavelengths (Molecular Devices).

EtBr accumulation assay

EtBr accumulation was measured as described earlier (Patkari et al. 2018). Briefly, *S. coelicolor* cells (WT86 and recombinants) were grown in R5 media till OD reached 1. The cells were further harvested and washed with 0.5 M chilled PBS at 12,000 rpm for 10 min at 4 °C. The cells were further diluted 100 times and 10⁶ cells were taken for

the assay. EtBr at a concentration of 5 µg/ml was added to the cells and accumulation was measured at 530 nm excitation and 585 nm emission wavelengths (Molecular Devices).

EtBr efflux assay

Briefly, *S. coelicolor* cells (WT86 and recombinants) were grown in R5 media till OD reached 1. Cells treated with EtBr alone and EtBr along with CCCP were incubated at 37°C for 60 min. Post incubation, cells were harvested and washed with 0.5 M chilled PBS at 12,000 rpm for 10 min at 4°C. Finally, cells were resuspended in chilled PBS. CCCP-treated cells were further saturated with CCCP and efflux was monitored over a period of 60 min by measuring the fluorescence at 530 nm excitation and 585 nm emission wavelengths (Molecular Devices).

Rifampicin uptake

S. coelicolor WT86 cells and recombinants were grown till 1 O.D and concentrated corresponding to OD 10. Rifampicin at MIC concentration (4 µg/ml) and 2X MIC concentration (8 µg/ml) was added to the cells and treatment was done for 1 h, post which cells were kept on ice and washed with chilled 0.5 M PBS at 12000 rpm for 5 min. The supernatant was taken to determine the extracellular rifampicin concentration by measuring absorbance at 274 nm through reverse-phase high-performance liquid chromatography (RP-HPLC) (Agilent). For HPLC, water acidified to pH 2.27 and acetonitrile were used in the ratio of 50:50 as mobile phase and passed through 10 mm C-18 column. Sample injection volume was kept at 20 µl and run time for each sample was kept for 5 min. Intracellular concentration was further derived by subtracting the extracellular concentration from initial added drug concentration. Final rifampicin uptake was calculated by normalization with the dry cell weight.

ROS measurement assay

Intracellular ROS in *S. coelicolor* WT86 and recombinants was measured using H₂-DCFDA (2, 7-dichlorofluorescein diacetate), a cell permeable fluorogenic dye. Upon diffusion of H₂-DCFDA inside the cells, it gets deacetylated into a non-fluorescent compound by the cellular esterases. This non-fluorescent compound then interacts with the intracellular ROS species to form a fluorescent compound 2,7-dichlorofluorescein which can be quantified (Eruslanov and Kusmartsev 2010). *S. coelicolor* WT86 cells along with the recombinants were grown till 1 O.D and treated with the desired conditions for 1 h. Furthermore, the cells were pelleted at 12,000 rpm for 10 min at 4°C and then resuspended in 0.5 M chilled PBS. Both untreated and treated cells were incubated with 10 µM of DCFDA for 15 min. Intracellular

ROS was measured by measuring the fluorescence at 435 nm excitation and 530 nm emission wavelengths. Ascorbate at a concentration of 400 µg/ml was used for the assay.

Cryo-field gun emission-scanning electron microscopy (Cryo-FEG-SEM)

S. coelicolor cells WT86 and the recombinants were grown till 1 O.D and washed with sterile MiliQ water at 5000 rpm for 10 min to wash off the media. This was followed by subsequently washing with different gradients of ethanol (20%, 10% and 5%). The cells were finally resuspended in 5–10 µl of water and visualized via the Cryo-FEG-SEM (JSM-7600F) platform. Scale bar and diameter of mycelia were calculated using the Image J software (Rueden et al. 2017).

Phylogenetic analysis

All sequences were retrieved from NCBI. Multiple sequence analysis was carried out using Clustal W algorithm with T-COFFE (Notredame et al. 2000). For highlighting the conserved amino acids, BOXSHADE was utilized (<http://sourceforge.net/projects/boxshade/>). Phylogenetic tree was constructed using the Neighbour Joining algorithm in the MEGA X software (Tamura et al. 2021). Reliability of the inner branches was carried out using bootstrap value of 1000 replicates.

Statistical analysis

All experiments were performed taking a minimum of three biological replicates. Significance was measured using Student's *T*-test with equal variances, where * indicates $p < 0.05$, ** indicates $p < 0.005$, *** indicates $p < 0.0005$.

Results

Conservation of SCO3366 in diverse *Streptomyces* and pathogens

SCO3366 has been recognized to encode an efflux pump belonging to MFS family of transporter proteins. It encodes a protein of 55 KDa and belongs to DHA2 group of transporter proteins with 14 transmembrane helices. To further unravel the evolutionary conservation of SCO3366 across *Streptomyces*, a BLASTp analysis was carried out which revealed SCO3366 to be widely conserved across 782 *Streptomyces* species, with a sequence coverage of >90% and identity >50%. These *Streptomyces* species were distributed among both soil and marine environments. A phylogenetic tree using 20 of these *Streptomyces* species was constructed using the Neighbour Joining algorithm, which highlights the

conservation of SCO3366 (Fig. 1). Apart from *Streptomyces*, homologs of SCO3366 were found to be distributed among different pathogenic mycobacteria and enterobacteria (Table S4). Furthermore, a Multiple Sequence Alignment of SCO3366 with transporters from these pathogens reveals the conservation of a large number of amino acid residues which predicts a common ancestral origin of these efflux pumps (Fig. S1).

SCO3366 imparts resistance to multiple antibiotics

To reveal the potential role of SCO3366 in antibiotic resistance, we created strains where native copy of the gene was deleted from WT *S. coelicolor* cells resulting in formation of Δ 3366 strain. Additionally, an overexpression strain, O3366, was constructed where multiple copies of *sco3366* were inserted into WT *S. coelicolor* cells by placing in pIJ86 plasmids under the *ermE***p* promoter. We also created the

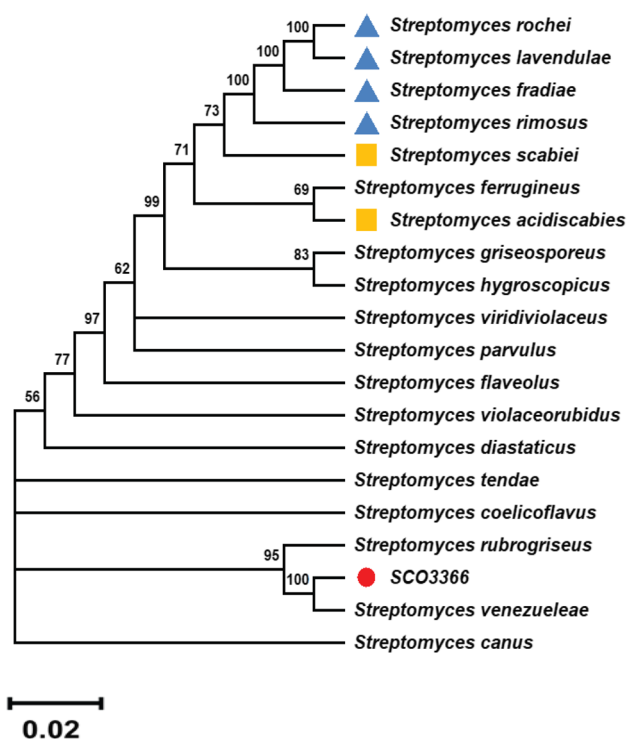


Fig. 1 Evolutionary conservation of SCO3366 across different *Streptomyces* species. Phylogenetic tree constructed using 20 *Streptomyces* species where homologs of SCO3366 were present with >90% sequence coverage and >50% identity. Blue represents *Streptomyces* from marine environments. Yellow represents pathogenic *Streptomyces*. All other represent *Streptomyces* from soil environment. The sequences were retrieved from NCBI. Sequences were aligned using the CLUSTAL W algorithm. Phylogenetic tree was constructed using Neighbour Joining algorithm in MEGA X. The data was created with bootstrap value of 1000 replicates. Scale bar of 0.02 represents estimated sequence divergence. The numbers on the nodes represent phylogenetic confidence of tree topology

complementation strain c Δ 3366 by inserting native copies of *sco3366* in Δ 3366 cells under pIJ86 plasmid. The susceptibility of O3366, Δ 3366 and c Δ 3366 towards different drugs was measured using the disc diffusion assay and compared to WT86 cells (cells which harboured only the empty pIJ86 plasmids). We found that O3366 provided increased resistance towards multiple drugs such as ciprofloxacin, a fluoroquinolone; chloramphenicol, an aminoglycoside; rifampicin, an RNA-synthesis inhibitor and the aromatic compound EtBr. This data is presented in Fig. 2. Interestingly, in the Δ 3366 strains, increased susceptibility was seen only towards rifampicin, suggesting that SCO3366 is a probable transporter of rifampicin. However, for the other drugs such as ciprofloxacin, chloramphenicol and EtBr, whereas overexpression of *sco3366* results in increased resistance, deletion of the gene did not lead to any change in the susceptibility profile, suggesting the prevalence of other efflux systems in *S. coelicolor* that also provide resistance towards these drugs. On the other hand, c Δ 3366 displayed similar resistance profile as that of O3366 strains towards ciprofloxacin, chloramphenicol and EtBr. However, a 2-fold decreased resistance was observed towards rifampicin in c Δ 3366 strains as compared to O3366 strains (Fig. 2). This can be attributed due to difference in the copies of *sco3366* gene in both the strains, thereby contributing to differential

resistance towards rifampicin. Note that, for each strain, the minimum concentration of the drug at which a clear visible zone was obtained corresponded to the MIC of the drug when measured via the agar dilution assay, thereby suggesting the two methods to be well correlated (Table 1).

Decreased intracellular drug levels in cells overexpressing SCO3366

To understand whether increased resistance is a consequence of a decrease in drug accumulation in O3366 cells, we measured the EtBr uptake profile in both O3366 and Δ 3366 cells. EtBr is a well-validated compound that is recognized as a substrate by a large number of efflux pumps and hence routinely used to understand the efflux-mediated bacterial export of drugs. In aqueous solutions, EtBr fluoresces weakly; however on entering the cells, the bound EtBr shows a significantly elevated fluorescence. The difference between the influx and efflux of the drugs out of the cells is a measure of the drug accumulated inside. We found that O3366 cells accumulated negligible EtBr levels as compared to WT86 cells, as confirmed by a decreased fluorescence intensity. Furthermore, blocking the efflux pathway via addition of an efflux inhibitor, CCCP, to O3366 cells, restored the

Fig. 2 Susceptibility of the *S. coelicolor* mutants O3366, Δ 3366 and c Δ 3366 to **a** ciprofloxacin, **b** chloramphenicol, **c** EtBr and **d** rifampicin. The disc diffusion assay is used to measure susceptibility to various drugs. Zone sizes are measured in millimetres. Dotted lines represent the minimum zone diameter that can be measured. Error bars indicate standard deviation calculated from three biological replicates. Statistical significance was calculated with respect to WT86 (*S. coelicolor* cells harbouring empty pIJ86 plasmids). *** indicates p -value <0.0005, ** indicates p -value <0.005, * indicates p -value <0.05, ns indicates no significance

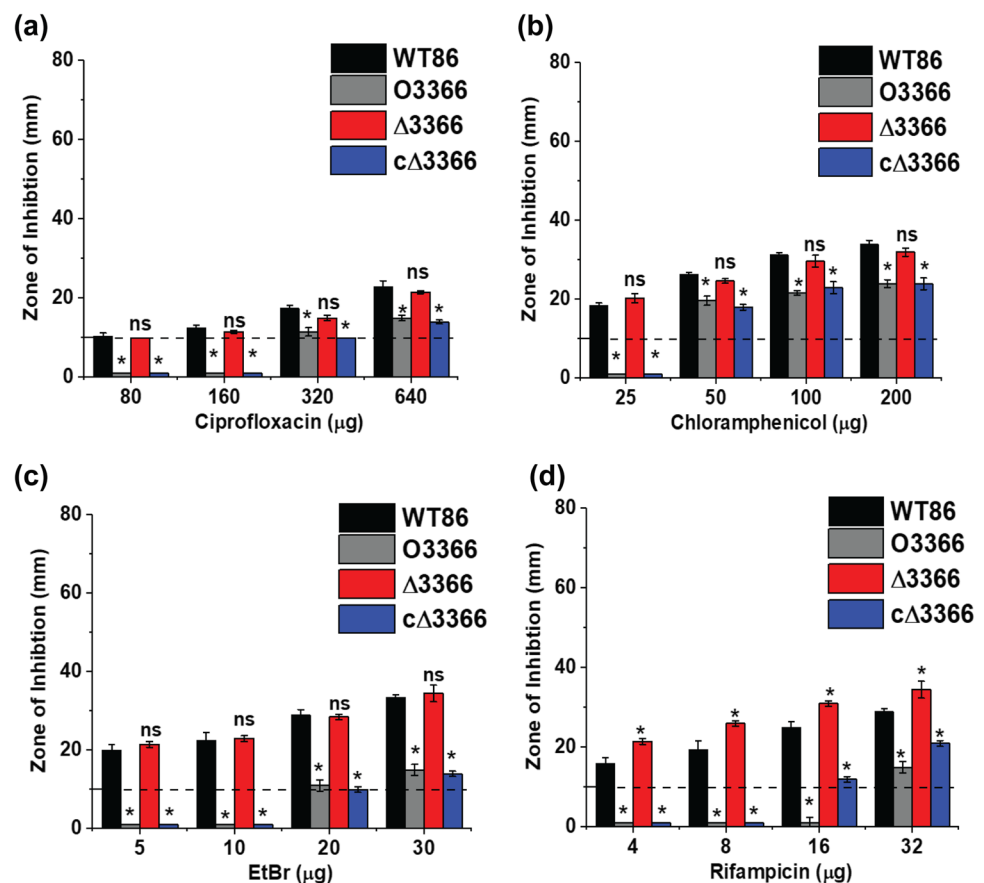


Table 1 Determination of minimum inhibitory concentration of different drugs against WT86, O3366 and Δ 3366 cells measured via agar dilution assay

Drug (μ g)	Minimum inhibitory concentration (μ g/ml)							
	WT86		O3366		Δ 3366		c Δ 3366	
	-CCCP	+CCCP	-CCCP	+CCCP	-CCCP	+CCCP	-CCCP	+CCCP
Ciprofloxacin	80	20	320	240	80	20	320	240
Chloramphenicol	20	5	40	20	20	5	40	20
Rifampicin	4	1	32	16	1	0.25	16	8
EtBr	2.5	0.5	20	2.5	2.5	0.5	20	2.5

EtBr levels to the WT86 cells (Fig. 3a). This data corresponds to the susceptibility profile, where addition of CCCP to O3366 cells restored the MIC of EtBr to WT86

levels (Table 1). Δ 3366 cells retained similar intracellular EtBr levels as that of WT86 cells, thus corresponding to the susceptibility profile towards the drug (Table 1). Upon

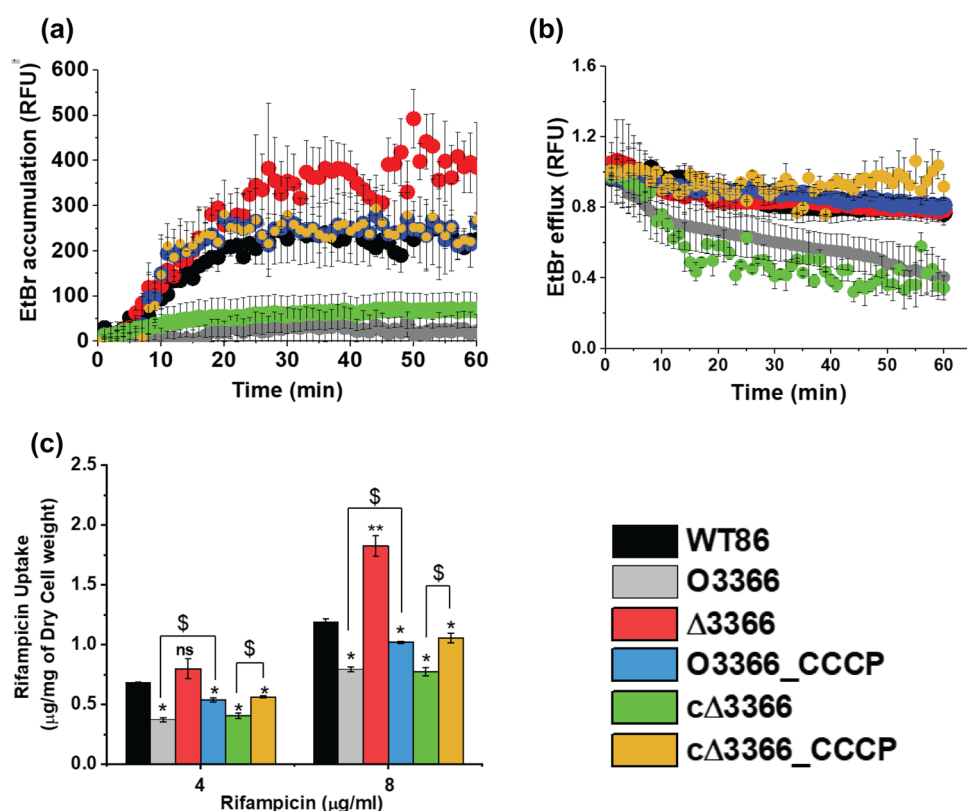


Fig. 3 Determination of intracellular drug accumulation in *S. coelicolor* WT86 and recombinants O3366, Δ 3366 and c Δ 3366. **a** Measurement of EtBr accumulation in O3366, Δ 3366 and c Δ 3366 cells with respect to WT86 cells. The accumulation was measured over a period of 60 min. **b** Determination of efflux activity of WT86 along with the recombinants O3366, Δ 3366 and c Δ 3366 cells measured over a period of 60 min. Cells were incubated with EtBr alone and EtBr along with CCCP for 60 min before measuring efflux. EtBr and CCCP were added at a final concentration of 5 μ g/ml respectively before initiation of both the assays. **c** Estimation of intracellular rifampicin in WT86, O3366, Δ 3366 and c Δ 3366 cells. Rifampicin levels were measured by treating the different strains at 4 μ g/ml (MIC concentration of WT86 cells) and 8 μ g/ml (2X MIC concentration of WT86 cells) respectively. Statistical significance of Δ 3366, O3366 and c Δ 3366 cells was calculated with respect to WT86 cells, treated

with respective rifampicin concentrations (shown with *). Accumulation of rifampicin in O3366 and c Δ 3366 cells was also measured upon addition of 5 μ g/ml CCCP to the rifampicin-treated cells and statistical significance in these cells was also calculated with respect to CCCP untreated cells (shown with \$). The colours black represent WT86, grey represent O3366, red represent Δ 3366 and green represent c Δ 3366. CCCP-treated O3366 and c Δ 3366 cells are shown in blue and yellow respectively. The legend displaying the colour codes for all strains used for drug accumulation assays is also shown. Error bars indicate standard deviation calculated from three independent biological replicates. 5 μ g/ml of CCCP was used in the assay. ** indicates p -value <0.005 and * indicates p -value <0.05 compared with respect to WT86 cells, \$ indicates p -value <0.05 compared with respect to CCCP untreated cells, ns indicates no significance

complementation of Δ 3366 cells with native copies of *sco3366* in c Δ 3366 cells, the accumulation of EtBr was similar to that in O3366 cells (Fig. 3a). Furthermore, addition of CCCP to these cells resulted in increased accumulation of EtBr, comparable to that of the WT86 cells. The corresponding susceptibility profile of c Δ 3366 cells towards EtBr in the absence and presence of CCCP was also found to be similar to O3366 cells. This data thus indicates that elevated resistance seen in O3366 and c Δ 3366 cells towards EtBr is due to decreased intracellular drug levels. Note that accumulation of EtBr in all the strains was measured with respect to the WT86 cells.

In order to further confirm that the decreased accumulation of EtBr in O3366 and c Δ 3366 cells is due to increased efflux and not due to limited intake of EtBr, we performed an EtBr efflux assay. This assay is a widely adopted technique for measurement of direct efflux in cells overexpressing efflux pumps (Blair et al. 2016). In this assay, WT86 along with the recombinants O3366, Δ 3366 and c Δ 3366 was preloaded with 2X MIC concentrations of EtBr (MIC corresponding to that of WT86 cells) and allowed to accumulate maximum amount of EtBr for 60 min. Cells were then washed to remove the extracellular EtBr and efflux was measured over a period of 60 min by recording the change in fluorescence intensity. We found that O3366 and c Δ 3366 cells were able to efflux EtBr faster than that of WT86 cells, whereas efflux activity in Δ 3366 cells was found to be similar to that of WT86 cells (Fig. 3b). Also, it was seen that O3366 and c Δ 3366 cells incubated with CCCP showed an efflux activity similar to that of WT86 cells, thereby suggesting the requirement of a proton gradient by SCO3366 to drive the efflux of EtBr. Hence from these results, we derive that increased resistance in O3366 and c Δ 3366 cells is a factor of EtBr efflux, thus resulting in elevated resistance towards the drug.

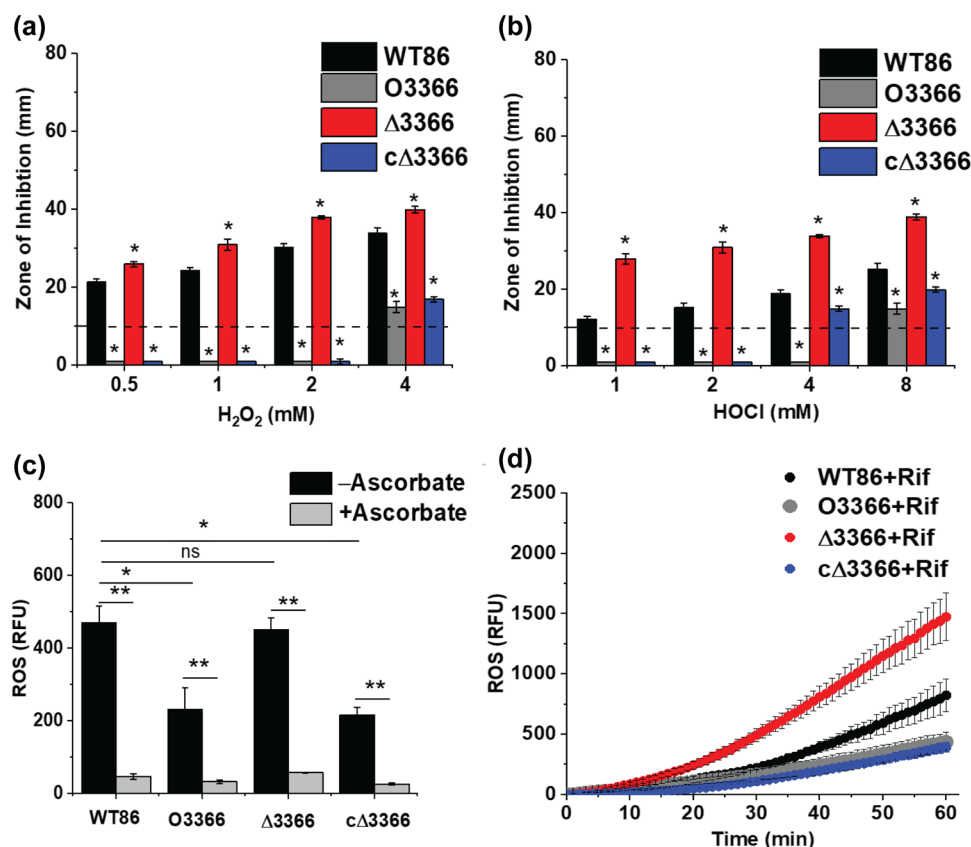
Furthermore, as rifampicin was found to be a major substrate of SCO3366, we measured the intracellular levels of rifampicin in O3366, Δ 3366 and c Δ 3366 cells compared to that in WT86 cells. The drug concentrations were chosen at MIC (4 μ g/ml) and 2X MIC (8 μ g/ml) with respect to the WT86 cells. The data is presented in Fig. 3c. We found that at both the drug concentrations, O3366 cells accumulated about ~2-fold decreased amounts of the drug as compared to the WT86 cells. On the other hand, while Δ 3366 cells accumulated nearly 2-fold elevated drug levels as compared to the WT86 cells at 2X MIC levels, the intracellular drug levels were similar at MIC drug concentration. Interestingly, upon addition of the efflux inhibitor CCCP at sub-inhibitory concentrations (5 μ g/ml), there was a significant increase in the intracellular drug concentration; however, the drug levels were not restored to WT86 levels. This data corresponded to the susceptibility data where on addition of CCCP, a 2-fold reduction in MIC of rifampicin was noted in O3366 cells; however, complete restoration of MIC to WT86 levels

was not seen (Table 1). Interestingly, c Δ 3366 cells showed a similar rifampicin uptake profile as that of O3366 cells, despite a 2-fold lower MIC towards the drug as compared to O3366 (Fig. 3c). Also, addition of CCCP to these cells led to an increase in intracellular rifampicin levels, however did not restore to WT86 levels (Fig. 3c). This data thus suggests that an additional factor of resistance, besides efflux, may perhaps be involved in providing elevated resistance to rifampicin.

Resistance to rifampicin is partially contributed by oxidative stress tolerance in O3366 cells

Oxidative stress is one of the major stresses a bacteria experiences in the course of its growth cycle (Ezraty et al. 2017; Imlay 2019). In order to gain tolerance to oxidative stress, bacteria produce several enzymes such as superoxide dismutase and catalase, which further help to quench the ROS species and maintain the redox balance in cells (Weydert and Cullen 2010). Interestingly, a few efflux pumps such as p55 in *Mycobacterium tuberculosis* and MacAB in *Salmonella typhimurium* have been found to serve a role in tolerance to oxidative stress, besides conferring resistance to multiple antibiotics (Bogomolnaya et al. 2013; Ramón-García et al. 2009). Using disc diffusion assay, we found that O3366 cells were able to provide increased tolerance to two potent oxidants in the environment, H₂O₂ and HOCl (Fig. 4a and b). Furthermore, deletion of *sco3366* led to increased susceptibility towards both the oxidants (Fig. 4a and b). However, on complementing the Δ 3366 strains with multiple copies of *sco3366* through the c Δ 3366 strain, the susceptibility profile towards both the oxidants was restored to that of O3366 cells (Fig. 4a and b). This data thus indicates SCO3366 to be essential in providing oxidative stress tolerance in *S. coelicolor*. We also quantified the basal ROS species in WT86, O3366, Δ 3366 and c Δ 3366 cells using the DCFDA assay. The basal ROS species generated in O3366 cells were nearly 3-fold lower as compared to the WT86 cells, Δ 3366 cells, displayed similar intracellular ROS levels as that of the WT86 cells (Fig. 4c). In the complementation strain, c Δ 3366, the ROS levels generated were similar to that observed in O3366 cells (Fig. 4c). The ROS levels generated in all the strains were further monitored in presence of an ROS quencher, ascorbate. Ascorbate acts a potential ROS quencher in different bacteria, which aids in survival in response to elevated oxidative stress (Dwyer et al. 2014). In presence of ascorbate, we observed nearly a 16-fold reduction of ROS species in WT86 and Δ 3366 cells, whereas in O3366 and c Δ 3366 cells, a further 6-fold reduction of ROS species was observed (Fig. 4c). Thus, from these results, we derive that apart from providing resistance to several

Fig. 4 Role of SCO3366 in oxidative stress tolerance. Susceptibility of the *S. coelicolor* mutants O3366 and Δ 3366 to **a** H_2O_2 and **b** HOCl. **c** Determination of basal intracellular ROS levels in WT86, O3366, Δ 3366 and c Δ 3366 cells in the absence and presence of sub-inhibitory concentrations of ascorbate (400 μ g/ml). **d** Determination of intracellular ROS measured in WT86 and recombinants O3366, Δ 3366 and c Δ 3366 post 60 min treatment with rifampicin. WT86 and Δ 3366 cells were treated with 4 μ g/ml of the drug; O3366 and c Δ 3366 were treated with 8 μ g/ml of the drug. ROS was measured using DCFDA assay. Plots for O3366 and c Δ 3366 are found to be overlapped. Statistical significance was calculated with respect to WT86 cells. Errors bars were calculated using three biological replicates. ** indicates p -value < 0.005, * indicates p -value < 0.05, ns indicates non-significant

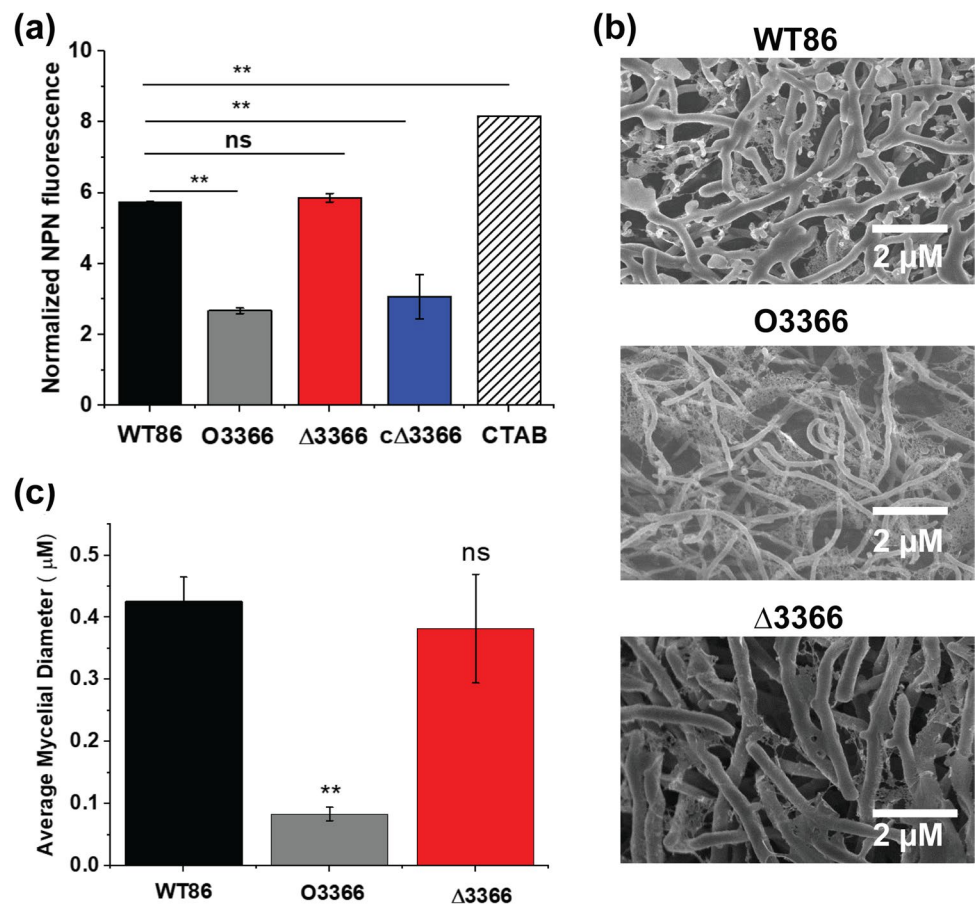


drugs, SCO3366 partly operates to quench the ROS species generated in vivo, thus maintaining redox homeostasis in the cells. Studies have shown that rifampicin leads to an outburst of ROS species such as hydroxyl and superoxide radicals, thus inducing cellular death in the bacteria such as in *Mycobacterium tuberculosis* (O'Sullivan et al. 2005; Piccaro et al. 2014; Shastri et al. 2018). Since SCO3366 was found to protect the bacteria against oxidative stress conditions, we quantified the intracellular ROS in WT86, O3366, Δ 3366 and c Δ 3366 cells in presence of rifampicin. For this assay, WT86 and Δ 3366 cells were treated with 4 μ g/ml of rifampicin, whereas O3366 and c Δ 3366 cells were treated with 8 μ g/ml of rifampicin. These concentrations were chosen because at these specific extracellular drug concentrations, the intracellular rifampicin levels were similar in the respective cells. We found that in the rifampicin-treated cells, intracellular ROS levels were lower in O3366 and c Δ 3366 cells as compared to the WT86 cells. Interestingly, in the Δ 3366 cells, a significant increase in intracellular ROS species was detected (Fig. 4d). This data therefore provides evidence that besides efflux, oxidative stress tolerance partly contributes to rifampicin resistance in cells overexpressing SCO3366, thus allowing survival under increased rifampicin concentrations.

Overexpression of SCO3366 results in decreased membrane permeability in *S. coelicolor*

Previous studies have identified a correlation between efflux pump overexpression and changes in membrane permeability. For example, the efflux pump EmhABC in *Pseudomonas fluorescens* cLP6a has been found to play a role in managing membrane permeability at higher incubation temperatures by modulating the fatty acid levels (Adebusuyi and Foght 2011). In a similar fashion, we further investigated whether SCO3366 has a role in governing membrane permeability. WT86 cells along with the recombinants O3366, Δ 3366 and c Δ 3366 cells were grown till mid-logarithmic phase and membrane permeability was monitored using the NPN assay. NPN is a hydrophobic dye which is excluded from the intact membrane, but when the membrane is permeabilized, NPN enters the cells, thus leading to increased fluorescence. This method has been widely used in determining membrane permeability in different Gram-positive and Gram-negative bacteria (Helander and Mattila-Sandholm 2000; Saito et al. 2019). We found that O3366 exhibit reduced NPN binding as compared to the WT86 cells, whereas the NPN binding profile of Δ 3366 cells was similar to that of WT86 cells (Fig. 5a). Complementation of Δ 3366

Fig. 5 Comparison of membrane permeability and cell morphology in different *S. coelicolor* strains. **a** Membrane integrity profile of *S. coelicolor* mutants O3366, Δ 3366 and c Δ 3366 measured using the NPN uptake assay. Data was plotted after normalization of NPN fluorescence with the CFU number. Statistical significance was calculated with respect to the WT86 cells. Cells treated with 30% CTAB were taken as the positive control. **b** Imaging of *S. coelicolor* WT86 cells along with recombinants O3366 and Δ 3366 under Cryo-FEG-SEM (JSM-7600F) platform. **c** Average mycelial diameter measured in WT86, O3366 and Δ 3366 cells. The diameter was calculated using the Image J software. Error bars indicate standard deviation calculated from three independent biological replicates. *** indicates p -value <0.005, ** indicates p -value <0.005, ns indicates no significance. Scale bars correspond to 2 μ M



with native copies of *sco3366* gene in c Δ 3366 resulted in restoration of NPN binding profile to that of O3366 cells. This data therefore highlights that increased expression of SCO3366 is associated with a decrease in membrane permeability. We noticed that the NPN binding profile of O3366 cells on CCCP treatment was similar to the untreated cells (Fig. S2), thus confirming the low NPN binding profile of O3366 cells is due to decreased permeability and not due to increased efflux of the dye.

Furthermore, a Cryo-FEG-SEM imaging was performed to understand the morphology of WT86 and the recombinants O3366 and Δ 3366 cells (Fig. 5b). O3366 cells showed a distinct phenotype compared to WT86 cells. The diameter of the mycelia was found to be 0.08 ± 0.01 μ M in O3366 cells as opposed to 0.4 ± 0.039 μ M that was noted in the case of WT86 cells. Δ 3366 cells, on the other hand, exhibit a similar phenotype as that of WT86 cells where diameter of mycelia corresponded to 0.39 ± 0.08 μ M (Fig. 5c). This data thus confirmed that overexpression of SCO3366 affects membrane permeability and could be a possible contributing factor of resistance to rifampicin.

sco3366 is transcriptionally regulated by antibiotics and oxidants

Our results so far demonstrate that overexpression of SCO3366 leads to elevated resistance to a broad spectrum of compounds that include several antibiotics and oxidants. In order to understand whether all these compounds are also involved in inducing the *sco3366* gene in the native host *Streptomyces coelicolor*, we monitored the expression of *sco3366* in the WT86 cells using quantitative real-time PCR assay. The induction of *sco3366* gene was measured in the presence of sub-inhibitory and inhibitory concentrations of all the compounds. We found that in presence of all the compounds tested, *sco3366* was upregulated in a concentration-dependent manner. In response to inhibitory concentration of the antibiotics ciprofloxacin, chloramphenicol and rifampicin (80 μ g/ml, 25 μ g/ml and 4 μ g/ml respectively), a 4-fold increased expression of *sco3366* was noted (Fig. 6a), whereas under inhibitory concentrations of the oxidants H_2O_2 (0.5 mM) and HOCl (1 mM), *sco3366* was upregulated to nearly 40-folds (Fig. 6b). This data thus suggests that *sco3366* is transcriptionally regulated by a wide range

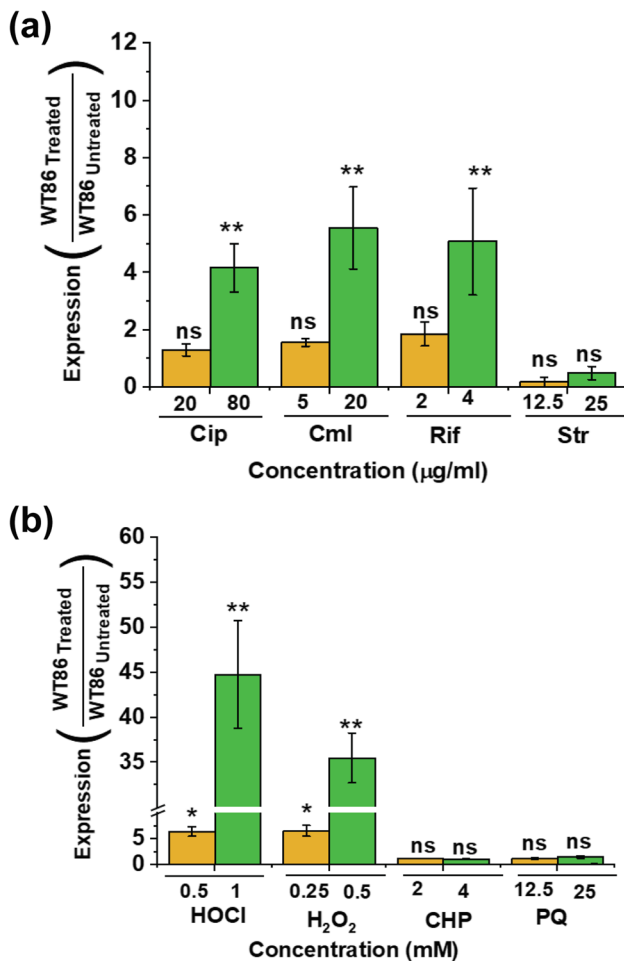


Fig. 6 Effect of antibiotics and oxidants on expression profile of *sco3366* gene in WT86 *S. coelicolor* cells. **a** Expression of *sco3366* under subinhibitory and inhibitory concentrations of ciprofloxacin (Cip), chloramphenicol (Cml), rifampicin (Rif) and ampicillin (Amp). **b** Expression of *sco3366* under subinhibitory and inhibitory concentrations of H₂O₂, HOCl (Hypochlorite), CHP (Cumene Hydroperoxide) and PQ (Paraquat). Treatment with the compounds was carried out for 1 h, followed by which expression was measured using real-time quantitative qPCR. Expression was normalized using 23S rRNA as the house-keeping gene, and further normalized to WT86 untreated cells. Statistical significance was calculated with respect to WT86 cells. Errors bars were calculated based on three biological replicates. *** indicates *p*-value <0.005, ** indicates *p*-value <0.005, ns indicates no significance

of compounds, which could either directly or indirectly lead to activation of *sco3366* gene under different environmental stress conditions. Note that overexpression or deletion of *sco3366* did not alter the susceptibility towards the antibiotic streptomycin and the oxidants cumene hydroperoxide and paraquat (data not shown). Also, these compounds were unable to induce the *sco3366* gene in the WT86 cells, and therefore were taken as negative controls in this assay. Also note that addition of apramycin, the antibiotic used for selection of pIJ86 plasmids, did not have any effect on the expression of *sco3366* in presence of different inducers (Fig. S3).

SCO3367 is involved in transcriptional regulation of *SCO3366*

Our studies so far describe *sco3366* to be transcriptionally governed by a variety of environmental stressors (Fig. 6). Efflux pumps are tightly regulated in bacteria, and expressed only in response to different stressors. SCO3366 was found to be placed adjacently and transcribing divergently from a TetR regulator, SCO3367. TetR regulators are tetrameric in nature and primarily classified as transcriptional repressors. However, in presence of different stressors, a TetR regulator changes its conformation in a way that is incompatible with DNA binding, thus allowing transcription of the respective gene (Kisker et al. 1995). Previous studies have revealed that SCO3367 binds to the *sco3366* promoter, thus inhibiting the transcription of the *sco3366* gene (Ahn et al. 2012). In our study, we further identified and evaluated whether regulation of SCO3366 is solely dependent on SCO3367, and also what are the conditions that lead to de-repression of *sco3366* gene. To answer these questions, firstly, we over-expressed *sco3367* by inserting multiple copies of the gene into WT *S. coelicolor* cells under the pIJ86 plasmid, thus yielding O3367 cells. Using real-time quantitative PCR, we found that *sco3366* gene was downregulated by nearly 9-folds in O3367 cells, thus confirming negative regulation of *sco3366* by SCO3367. Furthermore, since prior studies showed that *sco3366* was upregulated when treated in response to different antibiotics and oxidants recognized as SCO3366 substrates (Fig. 6), we further evaluated the ability of these inducers to relieve the repression via SCO3367. It was found that upon treatment of O3367 cells with inhibitory concentrations of the different inducers (Rifampicin-4μg/ml, H₂O₂-0.5 mM and HOCl-1 mM,) for 1 h, *sco3366* repression was partially relieved and became similar to basal levels; however, further upregulation was not observed (Fig. 7a). This is presumed to be due to the increased levels of SCO3367 repressor (16-folds), thus ensuing insufficient inactivation of the repressor in the 1-h incubation period. However, on further incubation of the cells with the drugs for 4 h, repression of *sco3366* promoter was relieved and increased expression of *sco3366* was noted (Fig. 7a). It was observed that nearly a 5-fold and 20-fold increased expression of *sco3366* was observed in presence of rifampicin and HOCl respectively, whereas a 40-fold upregulation of *sco3366* was noticed in response to H₂O₂. This confirms the role of SCO3367 in governing the expression of *sco3366* in presence of different stressors.

Furthermore, to identify whether SCO3367 directly regulates *sco3366* in presence of different inducers of *sco3366*, we studied the promoter activity of *sco3366* where SCO3367 was constitutively expressed. This was done by using a heterologous *Mycobacterium smegmatis* MC²155 host so as to negate the effect of any other regulatory factors. Constitutive

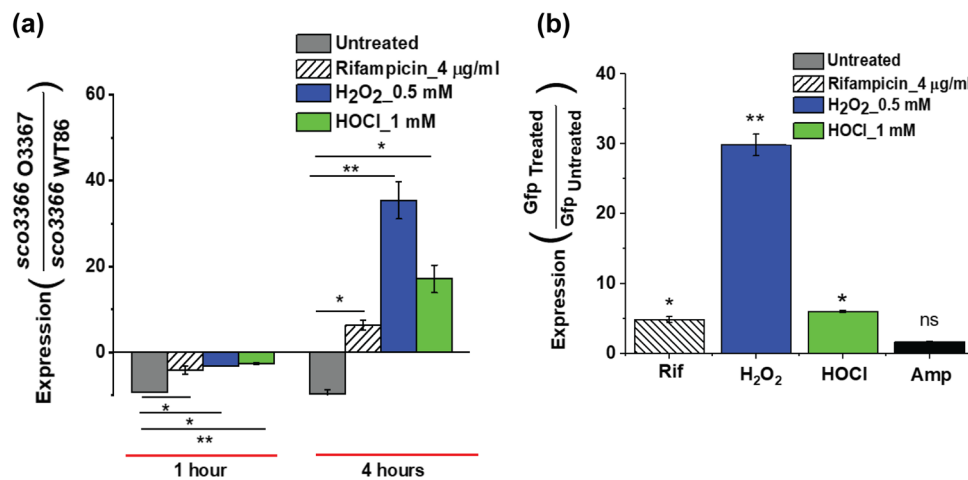


Fig. 7 Expression of *sco3366* in *S. coelicolor* and *M. smegmatis* recombinants. **a** Expression of *sco3366* in O3367 cells in the absence of any drug (untreated), treatment with inhibitory concentrations of rifampicin (4 µg/ml), H₂O₂ (0.5 mM) and HOCl (1mM). Expression was measured post 1 h and 4 h treatment with the drugs. Expression was measured using real-time quantitative PCR and normalized using 23S rRNA as the house-keeping gene, and further normalized to WT86 untreated cells. Statistical significance of O3367-treated cells was calculated with respect to O3367 untreated cells. **b** Expression of *gfp* in MS3367::P₃₃₆₆-gfp cells. Cells were treated with the

drugs at inhibitory concentration (Rifampicin-4 µg/ml, H₂O₂-0.25 mM, HOCl-1 mM) for 1 h before measuring *gfp* expression. Ampicillin was used as a negative control for the study. Expression was measured using real-time quantitative PCR method and fold change in expression was plotted. Negative fold change represents down-regulation of the specific gene. Expression was normalized using 16S rRNA as the house-keeping gene, and further normalized to untreated MS3367::P₃₃₆₆-gfp cells. Errors bars were calculated based on three biological replicates. ** represents *p*-value <0.005, * represents *p*-value <0.05

expression of SCO3367 in *M. smegmatis* was achieved by integrating the gene into mycobacterial genome via placing under the integrative plasmid pST-ki, thus yielding MS3367. Furthermore, the intergenic region between *sco3366*-*sco3367* was PCR amplified from the *S. coelicolor* genome and fused to a downstream *gfp* gene, thus forming P₃₃₆₆::gfp cells. These P₃₃₆₆::gfp cells were further transformed into competent MS3367 cells to form MS3367::P3366-gfp cells. Promoter activity of *sco3366* was monitored by measuring the *gfp* expression in these cells in the presence of inhibitory concentrations of rifampicin and the oxidants H₂O₂ and HOCl. *Pmyc* promoter with a downstream *gfp* was taken as the negative control. We found that in presence of inhibitory concentrations of rifampicin and HOCl, *sco3366* promoter was upregulated to 4-folds (Fig. 7b), whereas a 32-fold increased expression was noticed in presence of H₂O₂ (Fig. 7b). This data thus clearly indicates that SCO3367 directly regulates the expression of *sco3366*; and de-repression of *sco3366* gene occurs in response to diverse compounds which may either directly or indirectly interact with SCO3367, thus hindering the DNA binding activity of SCO3367 and leading to increased expression of *sco3366*. The beta-lactam drug ampicillin did not induce the expression of *sco3366* and therefore was taken as the negative control in this assay.

Discussion

In this study, we identified the role of a MFS transporter, SCO3366 from *Streptomyces coelicolor*, in conferring multi-drug resistance and tolerance to several physiological stressors. Phylogenetic analysis revealed homologs of SCO3366 to be ubiquitously distributed among different *Streptomyces* species from soil and marine environments, thus indicating the efflux pump to be a part of core genome that has evolved to serve a conserved functional role. Besides, homologs of SCO3366 were also observed in several pathogenic bacteria including different *Mycobacterium* species and entero-pathogens. Interestingly, a 40% similarity in the protein sequence of SCO3366 was observed with Rv1877, an integral membrane protein in *Mycobacterium tuberculosis*, which is a homolog of the widely characterized efflux pump LfrA from *Mycobacterium smegmatis* (Sharma and Bisht 2017).

Overexpression and knockout studies revealed the involvement of SCO3366 in providing resistance to multiple, structurally unrelated drugs, with rifampicin being the major substrate. Apart from multidrug resistance, SCO3366 was also involved in providing elevated tolerance to oxidative stress. Earlier studies have demonstrated several efflux pumps to be operational under oxidative stress, with resistance to antibiotics being a secondary effect (Bogomolnaya et al. 2013; Greene et al. 2018; Ramón-García et al. 2009). These efflux pumps extrude the toxic metabolites generated in response to various oxidants, thus leading to elevated

tolerance towards oxidative stress (Bogomolnaya et al. 2013; Ramón-García et al. 2009). Besides, these efflux pumps also aid in survival and virulence in different pathogenic bacteria dwelling in the macrophage environment (Pasqua et al. 2019). Interestingly, resistance to rifampicin has been shown to be partly contributed via quenching of ROS species in *Mycobacterium tuberculosis* (O'Sullivan et al. 2005). Thus, from our study, we derive that SCO3366 operates to protect the bacteria against oxidative stress and resistance towards rifampicin is partially contributed via tolerance of oxidative stress that is generated in response to drug.

Overexpression of SCO3366 also resulted in decreased membrane permeability in *S. coelicolor* with the diameter of mycelia reducing to nearly 5-folds of the WT86 cells. Earlier studies have indicated that overexpression of efflux pumps acts synergistically with decreased membrane permeability, thus retaining lower intracellular drug concentration and elevated resistance (Krishnamoorthy et al. 2017; Mallea et al. 1998). This phenomenon has been observed in the case of several clinical isolates of both Gram-positive and Gram-negative bacteria (Fernández and Hancock 2012). Role of transporters in governing cellular morphology and membrane permeability has been studied earlier. For example, the ABC transporter FtxABC has been found to be associated in regulation of cell wall architecture in the Gram-negative bacteria *E. coli* (Yang et al. 2011). Also, in *E. coli*, the efflux pump AcrAB has been found to modulate porin composition, thereby providing elevated resistance towards multiple drugs (Chollet et al. 2002). Our studies highlighted the role of SCO3366 to be essential in modulating membrane integrity in *S. coelicolor*. We also hypothesize that the decreased permeability resulting due to overexpression of SCO3366 is a possible contributing factor towards rifampicin resistance, by allowing decreased uptake of the drug.

Rifampicin is produced naturally by the bacteria *Amycolatopsis rifamycinica*, previously known as *Amycolatopsis (Streptomyces) mediterranei* (Saxena et al. 2014). However, a comparative analysis revealed SCO3366 to share a similarity of only 30% with the rifampicin specific transporters RifP and Ptr from *Amycolatopsis mediterranei* and *Streptomyces pristanspiralis* respectively (Blanc et al. 1995; Lei et al. 2018). Furthermore, comparison of data suggests SCO3366 to be more efficient in providing resistance towards rifampicin than the widely characterized Tap efflux system (BCG_1316c) from *Mycobacterium bovis* BCG, a close homolog of the Tap efflux pump (Rv_1258c) from *Mycobacterium tuberculosis* (Ramón-García et al. 2012). SCO3366 depicted an 8-fold increased resistance towards the drug upon being overexpressed into the native host as opposed to a 2-fold increased resistance seen in the case of Tap efflux pump. SCO3366 also showed a 27% similarity with the p55 efflux pump (Rv1410c) from *Mycobacterium tuberculosis*, which has been reported as the first efflux pump involved

in rifampicin export in mycobacteria (Ramón-García et al. 2009). These studies thus signify SCO3366 to be an important model in understanding the molecular mechanism of multi-substrate recognition and efflux, which will further help to shed light into the efflux-mediated resistance seen in different clinical pathogens.

Apart from providing resistance to rifampicin and oxidative stress, our data suggests that SCO3366 plays an essential physiological role in *Streptomyces coelicolor*. Overexpression of SCO3366 also resulted in increased export of the heavy metal iron (Fe), thereby providing increased tolerance towards the metal and accumulating negligible intracellular Fe levels. On the other hand, deletion of *sco3366* resulted in increased susceptibility towards Fe with a 3-fold elevated intracellular Fe levels (Fig. S4). Increased tolerance towards the organic solvent benzene was also observed in cells where SCO3366 was overexpressed, thus leading to increased survival when exposed to higher concentrations of the compound (Fig. S5). We also observed *sco3366* to be upregulated to nearly 25-folds in WT86 cells when exposed to inhibitory concentrations of FeSO₄ (400 µg/ml), whereas an upregulation of nearly 95-folds, 12-folds and 14-folds was noted in response to inhibitory concentrations of benzene (8%), toluene (16%) and ethanol (16%) respectively (Fig. S6). In addition to this, deletion of *sco3366* resulted in a sporulation defective phenotype when grown under minimal media such as mannitol soya agar plates as shown in Fig. S7a. No such sporulation defect was however noted when grown in rich media such as R5 plates (Fig. S7b). As the SCO3366-SCO3367 system is widely conserved in *Streptomyces* species, their role in governing physiology of *Streptomyces* species should be further investigated.

Furthermore, SCO3366 was found to be transcriptionally regulated by different compounds that were recognized as its substrates. Previous in vitro studies have identified an adjacently placed, divergently transcribing TetR regulator, SCO3367, to be involved in negative regulation of SCO3366 (Ahn et al. 2012). In our study, through in vivo approaches such as overexpression and promoter activity assays, we derived that SCO3367 can solely regulate the expression of SCO3366 without the aid of any additional regulators. The repression was further found to be relieved in the presence of different compounds that were recognized as substrates of SCO3366. Studies have shown that multiple substrates are able to induce transcriptional regulators through production of certain common cellular metabolites or through oxidative stress response, which leads to differential expression of several genes. This strategy thus helps the bacteria to survive in response to rapidly changing environments (Grove 2017; He et al. 2020). We therefore postulate a similar mechanism where a broad spectrum of compounds is found to relieve the repression of SCO3366 possibly via either a direct binding to the regulator or through a common signal generated in

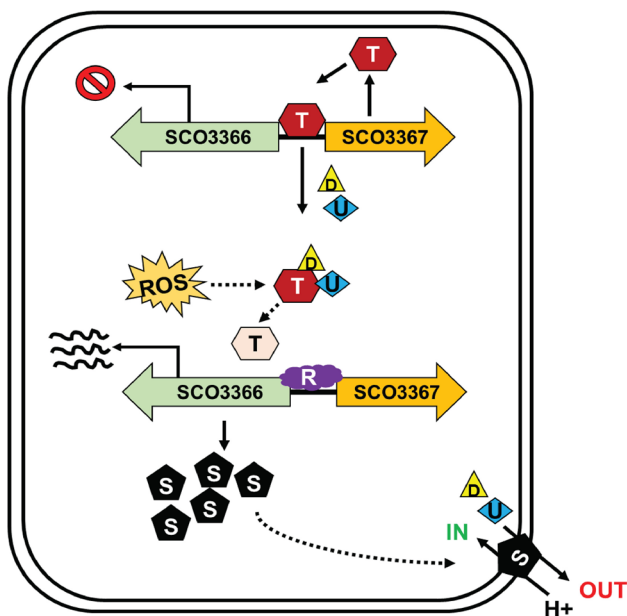


Fig. 8 Schematic illustration of regulation of SCO3366 by SCO3367. SCO3367 (T) binds strongly to the *sco3366* promoter thereby halting the expression of *sco3366* gene. In the presence of different substrates/ROS species, the repression of *sco3366* by SCO3367 is relieved, thus leading to binding of RNA polymerase (R) to the *sco3366* promoter and efficient transcription of SCO3366 efflux pump. SCO3366 protein (S) now localizes to the membrane and provides resistance to multiple drugs by utilizing energy from a proton-motive force

response to several inducers. Bioinformatics studies revealed SCO3367 to be conserved along with SCO3366 in 138 *Streptomyces* species from soil and marine environments thereby suggesting coevolution of the SCO3366-SCO3367 system to serve functional roles in protecting the bacteria under different environmental stressors. A phylogenetic tree depicting the conservation of SCO3366-SCO3367 system among different *Streptomyces* is presented in Fig. S8.

Thus, taken together, in this study, we present a novel efflux pump, SCO3366 from *Streptomyces coelicolor* which serves essential roles in eliciting multidrug resistance and protecting the bacteria against several physiological stressors. Our studies also highlighted the intricate mechanisms via which SCO3366 is regulated in *S. coelicolor*. We report that SCO3366 is repressed via SCO3367, and repression is relieved in response to different drugs which are recognized as substrates of SCO3366 (Fig. 8). Further studies deciphering the mechanism via which several drugs interact with SCO3367 will provide an insight into the molecular aspect of regulation of SCO3366. Since homologs of SCO3366 are present in different pathogenic bacteria, these studies can help in understanding the mode of regulation of efflux pumps occurring in different clinically important bacteria, and also aid in developing potential efflux inhibitors.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00253-022-11837-x>.

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Declarations

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Conflict of interest The authors declare no competing interests.

References

- Adebusuyi AA, Foght JM (2011) An alternative physiological role for the EmhABC efflux pump in *Pseudomonas fluorescens* cLP6a. BMC Microbiol 11(1):252. <https://doi.org/10.1186/1471-2180-11-252>
- Ahn SK, Cuthbertson L, Nodwell JR (2012) Genome context as a predictive tool for identifying regulatory targets of the TetR Family transcriptional regulators. PLoS ONE 7(11):e50562. <https://doi.org/10.1371/journal.pone.0050562>
- Blair JMA, Piddock LJV, Wright GD, Collier RJ (2016) How to measure export via bacterial multidrug resistance efflux pumps. mBio 7(4):e00840-16. <https://doi.org/10.1128/mBio.00840-16>
- Blanc V, Salah-Bey K, Folcher M, Thompson CJ (1995) Molecular characterization and transcriptional analysis of a multidrug resistance gene cloned from the pristinamycin-producing organism *Streptomyces Pristinamycin*. Mol Microbiol 17(5):989–999. https://doi.org/10.1111/j.1365-2958.1995.mmi_17050989.x
- Bogomolnaya LM, Andrews KD, Talamantes M, Maple A, Ragoza Y, Vazquez-Torres A, Andrews-Polymenis H (2013) The ABC-type efflux pump MacAB protects *Salmonella enterica* serovar typhimurium from oxidative stress. mBio 4(6):e00630-13. <https://doi.org/10.1128/mBio.00630-13>
- Boxshade: For multiple sequence analysis. Publisher. <http://sourceforge.net/projects/boxshade/>. Accessed January 5 2012
- Bradshaw E, Saalbach G, McArthur M (2013) Proteomic survey of the *Streptomyces coelicolor* nucleoid. J Proteomics 83:37–46. <https://doi.org/10.1016/j.jprot.2013.02.033>
- Chollet R, Bollet C, Chevalier J, Malléa M, Pagès J-M, Davin-Regli A (2002) *mar* operon involved in multidrug resistance of *Enterobacter aerogenes*. Antimicrob Agents Chemother 46(4):1093–1097. <https://doi.org/10.1128/AAC.46.4.1093-1097.2002>
- Costa SS, Sobkowiak B, Parreira R, Edgeworth JD, Viveiros M, Clark TG, Couto I (2019) Genetic diversity of *norA*, coding for a main efflux pump of *Staphylococcus aureus*. Front Genet 9(710) <https://doi.org/10.3389/fgene.2018.00710>
- D'Costa VM, McGrann KM, Hughes DW, Wright GD (2006) Sampling the antibiotic resistome. Science (new York, NY) 311(5759):374–377. <https://doi.org/10.1126/science.1120800>
- Dwyer DJ, Belenky PA, Yang JH, MacDonald IC, Martell JD, Takahashi N, Chan CTY, Lobritz MA, Braff D, Schwarz EG, Ye JD, Pati M, Vercruysse M, Ralifo PS, Allison KR, Khalil AS, Ting AY, Walker GC, Collins JJ (2014) Antibiotics induce redox-related physiological alterations as part of their lethality. Proc Natl Acad Sci U S A 111(20):E2100. <https://doi.org/10.1073/pnas.1401876111>
- Egan S, Wiener P, Kallifidas D, Wellington EMH (2001) Phylogeny of *Streptomyces* species and evidence for horizontal transfer of entire

- and partial antibiotic gene clusters. *Antonie Van Leeuwenhoek* 79(2):127–133. <https://doi.org/10.1023/A:1010296220929>
- Eruslanov E, Kusmartsev S (2010) Identification of ROS using oxidized DCFDA and flow-cytometry. *Methods Mol Biol* (clifton, NJ) 594:57–72. https://doi.org/10.1007/978-1-60761-411-1_4
- Ezraty B, Gennaris A, Barras F, Collet J-F (2017) Oxidative stress, protein damage and repair in bacteria. *Nat Rev Microbiol* 15(7):385–396. <https://doi.org/10.1038/nrmicro.2017.26>
- Fernández L, Hancock REW (2012) Adaptive and mutational resistance: role of porins and efflux pumps in drug resistance. *Clin Microbiol Rev* 25(4):661–681. <https://doi.org/10.1128/CMR.00043-12>
- Fraud S, Poole K (2011) Oxidative stress induction of the MexXY multidrug efflux genes and promotion of aminoglycoside resistance development in *Pseudomonas aeruginosa*. *Antimicrob Agents Chemother* 55(3):1068–1074. <https://doi.org/10.1128/AAC.01495-10>
- Greene NP, Kaplan E, Crow A, Koronakis V (2018) Antibiotic resistance mediated by the MacB ABC transporter family: a structural and functional perspective. *Front Microbiol* 9(950) <https://doi.org/10.3389/fmicb.2018.00950>
- Grove A (2017) Regulation of metabolic pathways by MarR family transcription factors. *Comput Struct Biotechnol J* 15:366–371. <https://doi.org/10.1016/j.csbj.2017.06.001>
- He H, Yuan S, Hu J, Chen J, Rang J, Tang J, Liu Z, Xia Z, Ding X, Hu S, Xia L (2020) Effect of the TetR family transcriptional regulator Sp1418 on the global metabolic network of *Saccharopolyspora pogona*. *Microb Cell Factories* 19(1):27. <https://doi.org/10.1186/s12934-020-01299-z>
- Helander IM, Mattila-Sandholm T (2000) Fluorometric assessment of Gram-negative bacterial permeabilization. *J Appl Microbiol* 88(2):213–219. <https://doi.org/10.1046/j.1365-2672.2000.00971.x>
- Hong HJ, Hutchings MI, Neu JM, Wright GD, Paget MS, Buttner MJ (2004) Characterization of an inducible vancomycin resistance system in *Streptomyces coelicolor* reveals a novel gene (vanK) required for drug resistance. *Mol Microbiol* 52(4):1107–1121. <https://doi.org/10.1111/j.1365-2958.2004.04032.x>
- Imlay JA (2019) Where in the world do bacteria experience oxidative stress? *Environ Microbiol* 21(2):521–530. <https://doi.org/10.1111/1462-2920.14445>
- Jiang X, Ellabaan MMH, Charusanti P, Munck C, Blin K, Tong Y, Weber T, Sommer MOA, Lee SY (2017) Dissemination of antibiotic resistance genes from antibiotic producers to pathogens. *Nat Commun* 8(1):15784. <https://doi.org/10.1038/ncomms15784>
- Jourdan S, Francis IM, Kim MJ, Salazar JJC, Planckaert S, Frère J-M, Matagne A, Kerff F, Devreese B, Loria R, Rigali S (2016) The CebE/MsiK transporter is a doorway to the cello-oligosaccharide-mediated induction of *Streptomyces scabies* pathogenicity. *Sci Rep* 6(1):27144. <https://doi.org/10.1038/srep27144>
- Kim YJ, Song JY, Hong SK, Smith CP, Chang YK (2008) Effects of pH shock on the secretion system in *Streptomyces coelicolor* A3(2). *J Microbiol Biotechnol* 18(4):658–662
- Kisker C, Hinrichs W, Tovar K, Hillen W, Saenger W (1995) The complex formed between Tet repressor and tetracycline-Mg²⁺ reveals mechanism of antibiotic resistance. *J Mol Biol* 247(2):260–280. <https://doi.org/10.1006/jmbi.1994.0138>
- Krishnamoorthy G, Leus IV, Weeks JW, Wolloscheck D, Rybenkov VV, Zgurskaya HI, Bonomo RA (2017) synergy between active efflux and outer membrane diffusion defines rules of antibiotic permeation into Gram-negative bacteria. *mBio* 8(5):e01172–1217. <https://doi.org/10.1128/mBio.01172-17>
- Lei C, Wang J, Liu Y, Liu X, Zhao G, Wang J (2018) A feedback regulatory model for RifQ-mediated repression of rifamycin export in *Amycolatopsis mediterranei*. *Microb Cell Factories* 17(1):14. <https://doi.org/10.1186/s12934-018-0863-5>
- Li L, Tetu SG, Paulsen IT, Hassan KA (2018) A transcriptomic approach to identify novel drug efflux pumps in bacteria. *Methods Mol Biol* (clifton, NJ) 1700:221–235. https://doi.org/10.1007/978-1-4939-7454-2_12
- Li W, Sharma M, Kaur P (2014) The DrrAB efflux system of *Streptomyces peuceetii* is a multidrug transporter of broad substrate specificity. *J Biol Chem* 289(18):12633–12646. <https://doi.org/10.1074/jbc.M113.536136>
- Li X-Z, Nikaido H (2009) Efflux-mediated drug resistance in bacteria: an update. *Drugs* 69(12):1555–1623. <https://doi.org/10.2165/11317030-000000000-00000>
- Lin J, Michel LO, Zhang Q (2002) CmeABC functions as a multidrug efflux system in *Campylobacter jejuni*. *Antimicrob Agents Chemother* 46(7):2124–2131. <https://doi.org/10.1128/aac.46.7.2124-2131.2002>
- Llanes C, Hocquet D, Vogne C, Benali-Baitich D, Neuwirth C, Plésiat P (2004) Clinical strains of *Pseudomonas aeruginosa* overproducing MexAB-OprM and MexXY efflux pumps simultaneously. *Antimicrob Agents Chemother* 48(5):1797–1802. <https://doi.org/10.1128/AAC.48.5.1797-1802.2004>
- Maarsingh JD, Yang S, Park JG, Haydel SE (2019) Comparative transcriptomics reveals PrrAB-mediated control of metabolic, respiration, energy-generating, and dormancy pathways in *Mycobacterium smegmatis*. *BMC Genom* 20(1):942. <https://doi.org/10.1186/s12864-019-6105-3>
- Machado D, Coelho TS, Perdigão J, Pereira C, Couto I, Portugal I, Maschmann RDA, Ramos DF, von Groll A, Rossetti MLR, Silva PA, Viveiros M (2017) Interplay between mutations and efflux in drug resistant clinical isolates of *Mycobacterium tuberculosis*. *Front Microbiol* 8(711) <https://doi.org/10.3389/fmicb.2017.00711>
- Mak S, Xu Y, Nodwell JR (2014) The expression of antibiotic resistance genes in antibiotic-producing bacteria. *Mol Microbiol* 93(3):391–402. <https://doi.org/10.1111/mmi.12689>
- Mallea M, Chevalier J, Bornet C, Eyraud A, Davin-Regli A, Bollet C, Pages JM (1998) Porin alteration and active efflux: two in vivo drug resistance strategies used by *Enterobacter aerogenes*. *Microbiology (reading, Engl)* 144(Pt 11):3003–3009. <https://doi.org/10.1099/00221287-144-11-3003>
- Nag A, Mehra S (2021) A Major Facilitator Superfamily (MFS) efflux pump, SCO4121, from *Streptomyces coelicolor* with roles in multidrug resistance and oxidative stress tolerance and its regulation by a MarR regulator. *Appl Environ Microbiol* 87(7) <https://doi.org/10.1128/aem.02238-20>
- Nikaido H, Zgurskaya HI (2001) AcrAB and related multidrug efflux pumps of *Escherichia coli*. *J Mol Microbiol Biotechnol* 3(2):215–218
- Notredame C, Higgins DG, Heringa J (2000) T-coffee: a novel method for fast and accurate multiple sequence alignment. *Edited by J Thornton. J Mol Biol* 302(1):205–217. <https://doi.org/10.1006/jmbi.2000.4042>
- O’Sullivan DM, McHugh TD, Gillespie SH (2005) Analysis of rpoB and pncA mutations in the published literature: an insight into the role of oxidative stress in *Mycobacterium tuberculosis* evolution. *J Antimicrob Chemother* 55(5):674–679. <https://doi.org/10.1093/jac/dki069>
- Pasqua M, Grossi M, Scinicariello S, Aussel L, Barras F, Colonna B, Prosseda G (2019) The MFS efflux pump EmrKY contributes to the survival of *Shigella* within macrophages. *Sci Rep* 9(1):2906. <https://doi.org/10.1038/s41598-019-39749-3>
- Patkari M, Kumbhar C, Nag A, Mehra S (2018) Distinct transcriptomic response of *S. coelicolor* to ciprofloxacin in a nutrient-rich environment. *Appl Microbiol Biotechnol* 102(24):10623–10643. <https://doi.org/10.1007/s00253-018-9398-2>

- Patkari M, Mehra S (2013) Transcriptomic study of ciprofloxacin resistance in *Streptomyces coelicolor* A3(2). *Mol Biosyst* 9(12):3101–3116. <https://doi.org/10.1039/c3mb70341j>
- Piccaro G, Pietraforte D, Giannoni F, Mustazzolu A, Fattorini L (2014) Rifampin induces hydroxyl radical formation in *Mycobacterium tuberculosis*. *Antimicrob Agents Chemother* 58(12):7527–7533. <https://doi.org/10.1128/AAC.03169-14>
- Piddock LJV (2006) Multidrug-resistance efflux pumps not just for resistance. *Nat Rev Microbiol* 4(8):629–636. <https://doi.org/10.1038/nrmicro1464>
- Piddock LJV, Johnson MM, Simjee S, Pumbwe L (2002) Expression of efflux pump gene *pmrA* in fluoroquinolone-resistant and -susceptible clinical isolates of *Streptococcus pneumoniae*. *Antimicrob Agents Chemother* 46(3):808–812. <https://doi.org/10.1128/AAC.46.3.808-812.2002>
- Ramón-García S, Martín C, Thompson CJ, Aínsa JA (2009) Role of the *Mycobacterium tuberculosis* P55 efflux pump in intrinsic drug resistance, oxidative stress responses, and growth. *Antimicrob Agents Chemother* 53(9):3675–3682. <https://doi.org/10.1128/aac.00550-09>
- Ramón-García S, Mick V, Dainese E, Martín C, Thompson CJ, De Rossi E, Manganelli R, Aínsa JA (2012) Functional and genetic characterization of the tap efflux pump in *Mycobacterium bovis* BCG. *Antimicrob Agents Chemother* 56(4):2074–2083. <https://doi.org/10.1128/AAC.05946-11>
- Rodríguez-Herva JJ, García V, Hurtado A, Segura A, Ramos JL (2007) The ttgGHI solvent efflux pump operon of *Pseudomonas putida* DOT-T1E is located on a large self-transmissible plasmid. *Environ Microbiol* 9(6):1550–1561. <https://doi.org/10.1111/j.1462-2920.2007.01276.x>
- Rueden CT, Schindelin J, Hiner MC, DeZonia BE, Walter AE, Arena ET, Eliceiri KW (2017) ImageJ2: ImageJ for the next generation of scientific image data. *BMC Bioinform* 18(1):529–529. <https://doi.org/10.1186/s12859-017-1934-z>
- Saier MH Jr, Paulsen IT (2001) Phylogeny of multidrug transporters. *Semin Cell Dev Biol* 12(3):205–213. <https://doi.org/10.1006/scdb.2000.0246>
- Saito H, Sakakibara Y, Sakata A, Kurashige R, Murakami D, Kageshima H, Saito A, Miyazaki Y (2019) Antibacterial activity of lysozyme-chitosan oligosaccharide conjugates (LYZOX) against *Pseudomonas aeruginosa*, *Acinetobacter baumannii* and Methicillin-resistant *Staphylococcus aureus*. *PLoS ONE* 14(5):e0217504. <https://doi.org/10.1371/journal.pone.0217504>
- Saxena A, Kumari R, Mukherjee U, Singh P, Lal R (2014) Draft genome sequence of the rifamycin producer *Amycolatopsis rifamycinica* DSM 46095. *Genome Announc* 2(4):e00662–e714. <https://doi.org/10.1128/genomeA.00662-14>
- Schmittgen TD, Livak KJ (2008) Analyzing real-time PCR data by the comparative CT method. *Nat Protoc* 3(6):1101–1108. <https://doi.org/10.1038/nprot.2008.73>
- Schneider C, Mack M (2021) A second riboflavin import system is present in flavinogenic *Streptomyces davaonensis* and supports roseoflavin biosynthesis. *Mol Microbiol* 2:470–482. <https://doi.org/10.1111/mmi.14726>
- Sharma D, Bisht D (2017) Role of Bacterioferritin & Ferritin in *M. tuberculosis* pathogenesis and drug resistance: a future perspective by interactomic approach. *Front Cell Infect Microbiol* 7(240) <https://doi.org/10.3389/fcimb.2017.00240>
- Shastri MD, Shukla SD, Chong WC, Dua K, Peterson GM, Patel RP, Hansbro PM, Eri R, O'Toole RF (2018) Role of oxidative stress in the pathology and management of human tuberculosis. *Oxid Med Cell Longev* 2018:7695364. <https://doi.org/10.1155/2018/7695364>
- Tamura K, Stecher G, Kumar S (2021) MEGA11: molecular evolutionary genetics analysis version 11. *Mol Biol Evol* 38(7):3022–3027. <https://doi.org/10.1093/molbev/msab120>
- Van Dessel W, Van Mellaert L, Geukens N, Lammertyn E, Anné J (2004) Isolation of high quality RNA from *Streptomyces*. *J Microbiol Methods* 58(1):135–137. <https://doi.org/10.1016/j.mimet.2004.03.015>
- Vecchione JJ, Alexander B Jr, Sello JK (2009) Two distinct major facilitator superfamily drug efflux pumps mediate chloramphenicol resistance in *Streptomyces coelicolor*. *Antimicrob Agents Chemother* 53(11):4673–4677. <https://doi.org/10.1128/AAC.00853-09>
- Wang T-J, Shan Y-M, Li H, Dou W-W, Jiang X-H, Mao X-M, Liu S-P, Guan W-J, Li Y-Q (2017) Multiple transporters are involved in natamycin efflux in *Streptomyces chattanoogensis* L10. *Mol Microbiol* 103(4):713–728. <https://doi.org/10.1111/mmi.13583>
- Webber MA, Piddock LJV (2003) The importance of efflux pumps in bacterial antibiotic resistance. *J Antimicrob Chemother* 51(1):9–11. <https://doi.org/10.1093/jac/dkg050>
- Weydert CJ, Cullen JJ (2010) Measurement of superoxide dismutase, catalase and glutathione peroxidase in cultured cells and tissue. *Nat Protoc* 5(1):51–66. <https://doi.org/10.1038/nprot.2009.197>
- Yang DC, Peters NT, Parzych KR, Uehara T, Markovski M, Bernhardt TG (2011) An ATP-binding cassette transporter-like complex governs cell-wall hydrolysis at the bacterial cytoskeletal ring. *Proc Natl Acad Sci U S A* 108(45):18209
- Yu L, Yan X, Wang L, Chu J, Zhuang Y, Zhang S, Guo M (2012) Molecular cloning and functional characterization of an ATP-binding cassette transporter OtrC from *Streptomyces rimosus*. *BMC Biotechnol* 12:52. <https://doi.org/10.1186/1472-6750-12-52>
- Zgurskaya HI, Rybenkov VV, Krishnamoorthy G, Leus IV (2018) Trans-envelope multidrug efflux pumps of Gram-negative bacteria and their synergism with the outer membrane barrier. *Res Microbiol* 169(7–8):351–356. <https://doi.org/10.1016/j.resmic.2018.02.002>
- Zhou Z, Sun N, Wu S, Li Y-Q, Wang Y (2016) Genomic data mining reveals a rich repertoire of transport proteins in *Streptomyces*. *BMC Genom* 17 Suppl 7(Suppl 7):510–510. <https://doi.org/10.1186/s12864-016-2899-4>

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